

# Explicit Derivation of Kinematical and Simplified Two-Beam Dynamical Rocking Curves with Absorptive Extensions

## Definitions and conventions

We define

$$t = \text{crystal thickness}, \quad s = \text{excitation error}, \quad \xi = \text{extinction distance},$$

and the dimensionless quantity

$$w = \xi s.$$

Here,

- $s = 0$  corresponds to the exact Bragg condition,
- $s$  has units of inverse length,
- $\xi$  has units of length,
- therefore both  $st$  and  $\xi s$  are dimensionless.

We define the normalized sinc function by

$$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}. \quad (1)$$

Throughout,  $u_0(z)$  denotes the transmitted-beam amplitude and  $u_{\mathbf{g}}(z)$  the diffracted-beam amplitude for reflection  $\mathbf{g}$ .

The purpose of this note is to derive:

1. the kinematical rocking curve,
2. the simplified two-beam dynamical rocking curve,
3. a simple absorptive extension of the dynamical rocking curve, and
4. a phenomenological comparison of the absorption scales used in the kinematical and dynamical treatments.

## 1 Background: two-beam wavefield and coupled equations

If only the transmitted beam and one diffracted beam are retained, the crystal wavefield is written in the two-beam form

$$\psi(\mathbf{r}) = u_0(z)e^{2\pi i \mathbf{k} \cdot \mathbf{r}} + u_{\mathbf{g}}(z)e^{2\pi i (\mathbf{k} + \mathbf{g}) \cdot \mathbf{r}}.$$

Here the exponential factors describe the fast oscillations, while the amplitudes  $u_0(z)$  and  $u_{\mathbf{g}}(z)$  vary with depth  $z$ .

We substitute this ansatz into a Schrödinger-type crystal wave equation

$$[\nabla^2 + K^2] \psi(\mathbf{r}) = V(\mathbf{r})\psi(\mathbf{r}), \quad V(\mathbf{r}) = \sum_{\mathbf{h}} V_{\mathbf{h}} e^{2\pi i \mathbf{h} \cdot \mathbf{r}}.$$

## Apply the Laplacian to each beam

For a term of the form

$$u(z)e^{2\pi i \mathbf{q} \cdot \mathbf{r}},$$

with  $\mathbf{q}$  constant and  $u$  depending only on  $z$ ,

$$\nabla^2 [u(z)e^{2\pi i \mathbf{q} \cdot \mathbf{r}}] = e^{2\pi i \mathbf{q} \cdot \mathbf{r}} (u''(z) + 4\pi i q_z u'(z) - 4\pi^2 |\mathbf{q}|^2 u(z)).$$

Applying this to the two retained beams gives

$$\nabla^2 [u_0 e^{2\pi i \mathbf{k} \cdot \mathbf{r}}] = e^{2\pi i \mathbf{k} \cdot \mathbf{r}} (u_0'' + 4\pi i k_z u_0' - 4\pi^2 |\mathbf{k}|^2 u_0),$$

and

$$\nabla^2 [u_{\mathbf{g}} e^{2\pi i (\mathbf{k} + \mathbf{g}) \cdot \mathbf{r}}] = e^{2\pi i (\mathbf{k} + \mathbf{g}) \cdot \mathbf{r}} (u_{\mathbf{g}}'' + 4\pi i (k_z + g_z) u_{\mathbf{g}}' - 4\pi^2 |\mathbf{k} + \mathbf{g}|^2 u_{\mathbf{g}}).$$

Therefore,

$$(\nabla^2 + K^2)\psi = e^{2\pi i \mathbf{k} \cdot \mathbf{r}} [u_0'' + 4\pi i k_z u_0' + (K^2 - 4\pi^2 |\mathbf{k}|^2) u_0] + e^{2\pi i (\mathbf{k} + \mathbf{g}) \cdot \mathbf{r}} [u_{\mathbf{g}}'' + 4\pi i (k_z + g_z) u_{\mathbf{g}}' + (K^2 - 4\pi^2 |\mathbf{k} + \mathbf{g}|^2) u_{\mathbf{g}}]. \quad (2)$$

## Expand the potential term and collect equal phases

Now multiply the potential by  $\psi$ :

$$V(\mathbf{r})\psi(\mathbf{r}) = \left( \sum_{\mathbf{h}} V_{\mathbf{h}} e^{2\pi i \mathbf{h} \cdot \mathbf{r}} \right) [u_0 e^{2\pi i \mathbf{k} \cdot \mathbf{r}} + u_{\mathbf{g}} e^{2\pi i (\mathbf{k} + \mathbf{g}) \cdot \mathbf{r}}].$$

This produces terms of the form

$$V_{\mathbf{h}} u_0 e^{2\pi i (\mathbf{k} + \mathbf{h}) \cdot \mathbf{r}}, \quad V_{\mathbf{h}} u_{\mathbf{g}} e^{2\pi i (\mathbf{k} + \mathbf{g} + \mathbf{h}) \cdot \mathbf{r}}.$$

To match the phase  $e^{2\pi i \mathbf{k} \cdot \mathbf{r}}$ , one must have

$$\mathbf{h} = 0 \quad \text{from the } u_0 \text{ term}, \quad \mathbf{h} = -\mathbf{g} \quad \text{from the } u_{\mathbf{g}} \text{ term}.$$

To match the phase  $e^{2\pi i (\mathbf{k} + \mathbf{g}) \cdot \mathbf{r}}$ , one must have

$$\mathbf{h} = \mathbf{g} \quad \text{from the } u_0 \text{ term}, \quad \mathbf{h} = 0 \quad \text{from the } u_{\mathbf{g}} \text{ term}.$$

Thus, in the two-beam approximation, the only retained couplings are:

- $V_0$  acting on each beam,
- $V_{\mathbf{g}}$  coupling  $u_0 \rightarrow u_{\mathbf{g}}$ ,
- $V_{-\mathbf{g}}$  coupling  $u_{\mathbf{g}} \rightarrow u_0$ .

This is exactly why the problem reduces to a  $2 \times 2$  coupled system.

Collecting coefficients of the two retained phase factors gives

$$u_0'' + 4\pi i k_z u_0' + (K^2 - 4\pi^2 |\mathbf{k}|^2) u_0 = V_0 u_0 + V_{-\mathbf{g}} u_{\mathbf{g}},$$

$$u_{\mathbf{g}}'' + 4\pi i (k_z + g_z) u_{\mathbf{g}}' + (K^2 - 4\pi^2 |\mathbf{k} + \mathbf{g}|^2) u_{\mathbf{g}} = V_{\mathbf{g}} u_0 + V_0 u_{\mathbf{g}}.$$

## Slowly varying envelope approximation

Assuming that the amplitudes vary slowly with depth, one neglects the second derivatives:

$$u_0'' \approx 0, \quad u_{\mathbf{g}}'' \approx 0.$$

After the standard rearrangement of common phase terms and absorption of constants into the conventional definitions of excitation error  $s$  and extinction distance  $\xi$ , the equations can be written as

$$\frac{du_0}{dz} = i\pi s u_0 + i\pi \frac{1}{\xi} u_{\mathbf{g}}, \quad (3)$$

$$\frac{du_{\mathbf{g}}}{dz} = i\pi \frac{1}{\xi} u_0 - i\pi s u_{\mathbf{g}}. \quad (4)$$

In matrix form,

$$\frac{d}{dz} \begin{pmatrix} u_0 \\ u_{\mathbf{g}} \end{pmatrix} = i\pi \begin{pmatrix} s & 1/\xi \\ 1/\xi & -s \end{pmatrix} \begin{pmatrix} u_0 \\ u_{\mathbf{g}} \end{pmatrix}.$$

These same amplitudes  $u_0(z)$  and  $u_{\mathbf{g}}(z)$  are used in both the kinematical and dynamical treatments below. In the kinematical limit, one assumes  $u_0(z)$  remains approximately constant and derives a single thickness integral. The full derivation of that result is given in Section 2. The kinematical and dynamical sections may be written in slightly different phase conventions; in the comparison below they are placed on a common normalized-sinc footing, and the local repair is only the overall prefactor consistency.

## 2 Kinematical rocking curve

### 2.1 Physical assumption

In the kinematical approximation, scattering into the diffracted beam is assumed to be weak, so that the transmitted beam is not significantly depleted as it propagates through the crystal. One therefore approximates

$$u_0(z) \approx \text{constant},$$

and with the usual normalization takes

$$u_0(z) \approx 1.$$

Under this assumption, the coupling back from the diffracted beam into the transmitted beam is neglected, and the diffracted amplitude is obtained from the diffracted-beam equation alone. Physically, this corresponds to single scattering.

### 2.2 Thickness integral from the same starting point

Before the symmetric phase convention leading to Eqs. (3) and (4) is imposed, the diffracted-beam equation can be written schematically as

$$\frac{du_{\mathbf{g}}}{dz} - 2\pi i s u_{\mathbf{g}} = C u_0,$$

where  $C$  is a coupling constant proportional to  $1/\xi$ .

Making the kinematical approximation  $u_0(z) \approx 1$  gives

$$\frac{du_{\mathbf{g}}}{dz} - 2\pi i s u_{\mathbf{g}} = C.$$

This is a first-order linear inhomogeneous differential equation. Using the integrating factor

$$e^{-2\pi isz},$$

one obtains

$$\frac{d}{dz} (u_{\mathbf{g}}(z)e^{-2\pi isz}) = Ce^{-2\pi isz}.$$

Integrating from  $z = 0$  to  $z = t$ , with the entrance condition  $u_{\mathbf{g}}(0) = 0$ , yields

$$u_{\mathbf{g}}(t)e^{-2\pi ist} = C \int_0^t e^{-2\pi isz} dz.$$

Hence

$$u_{\mathbf{g}}(t) = Ce^{2\pi ist} \int_0^t e^{-2\pi isz} dz.$$

Up to an overall phase factor, this is equivalent to

$$u_{\mathbf{g}}^{(\text{kin})}(t, s) \propto \frac{1}{\xi} \int_0^t e^{2\pi isz} dz.$$

Thus the usual kinematical thickness integral follows directly from the same two-beam starting point once the undepleted-beam approximation is made.

### 2.3 Evaluation of the integral

We now evaluate

$$\int_0^t e^{2\pi isz} dz.$$

Since

$$\int e^{2\pi isz} dz = \frac{1}{2\pi is} e^{2\pi isz},$$

it follows that

$$u_{\mathbf{g}}^{(\text{kin})}(t, s) \propto \frac{1}{\xi} \frac{e^{2\pi ist} - 1}{2\pi is}.$$

Rewrite the numerator as

$$e^{2\pi ist} - 1 = e^{\pi ist} (e^{\pi ist} - e^{-\pi ist}) = e^{\pi ist} 2i \sin(\pi st).$$

Therefore,

$$u_{\mathbf{g}}^{(\text{kin})}(t, s) \propto \frac{1}{\xi} e^{\pi ist} \frac{\sin(\pi st)}{\pi s}.$$

Multiplying and dividing by  $t$  gives

$$u_{\mathbf{g}}^{(\text{kin})}(t, s) = \frac{t}{\xi} e^{\pi ist} \frac{\sin(\pi st)}{\pi st} = \frac{t}{\xi} e^{\pi ist} \text{sinc}(st).$$

### 2.4 Kinematical intensity

The diffracted intensity is proportional to the squared modulus of the amplitude:

$$I_{\text{kin}}(s) \propto \left| u_{\mathbf{g}}^{(\text{kin})}(t, s) \right|^2.$$

Since  $|e^{\pi ist}|^2 = 1$ , and under the same normalized-sinc prefactor convention used for comparison with the dynamical sinc rewrite, one obtains

$$I_{\text{kin}}(s) \propto \pi^2 \left( \frac{t}{\xi} \right)^2 \text{sinc}^2(st).$$

Using  $w = \xi s$ , this may also be written as

$$I_{\text{kin}}(w) \propto \pi^2 \left( \frac{t}{\xi} \right)^2 \text{sinc}^2\left(\frac{t}{\xi} w\right).$$

### 3 Simplified two-beam dynamical derivation

#### 3.1 Why the dynamical case is different

In the kinematical derivation, the transmitted beam was assumed to remain effectively unchanged, so the diffracted amplitude could be written as a single coherent thickness integral.

In the dynamical case, this approximation is no longer valid. The transmitted and diffracted beams exchange amplitude continuously as they propagate through the crystal. One must therefore solve the coupled system in Eqs. (3) and (4), rather than evaluate a single integral.

#### 3.2 Two-beam coupled system

We start from

$$\frac{d}{dz} \begin{pmatrix} u_0(z) \\ u_{\mathbf{g}}(z) \end{pmatrix} = i\pi \begin{pmatrix} s & 1/\xi \\ 1/\xi & -s \end{pmatrix} \begin{pmatrix} u_0(z) \\ u_{\mathbf{g}}(z) \end{pmatrix},$$

with entrance condition

$$u_0(0) = 1, \quad u_{\mathbf{g}}(0) = 0.$$

Define

$$\mathbf{u}(z) = \begin{pmatrix} u_0(z) \\ u_{\mathbf{g}}(z) \end{pmatrix}, \quad M = \begin{pmatrix} s & 1/\xi \\ 1/\xi & -s \end{pmatrix}.$$

Then

$$\frac{d\mathbf{u}}{dz} = i\pi M \mathbf{u}.$$

The formal solution is

$$\mathbf{u}(z) = e^{i\pi z M} \mathbf{u}(0).$$

So the problem reduces to evaluating the matrix exponential. We compute  $M^2$

$$M^2 = \begin{pmatrix} s & 1/\xi \\ 1/\xi & -s \end{pmatrix} \begin{pmatrix} s & 1/\xi \\ 1/\xi & -s \end{pmatrix} = \left(s^2 + \frac{1}{\xi^2}\right) I,$$

and define

$$\Omega = \sqrt{s^2 + \frac{1}{\xi^2}}.$$

Then

$$M^2 = \Omega^2 I.$$

#### 3.3 Evaluate the matrix exponential

Using the power-series expansion of the exponential,

$$e^{i\pi z M} = \sum_{n=0}^{\infty} \frac{(i\pi z M)^n}{n!},$$

and the relation  $M^2 = \Omega^2 I$ , the even and odd powers of  $M$  can be grouped separately:

$$M^{2k} = \Omega^{2k} I, \quad M^{2k+1} = \Omega^{2k} M.$$

Therefore,

$$e^{i\pi z M} = I \cos(\pi z \Omega) + i \frac{\sin(\pi z \Omega)}{\Omega} M.$$

### 3.4 Apply the entrance condition

Applying this to

$$\mathbf{u}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

gives

$$\mathbf{u}(z) = \left[ I \cos(\pi z \Omega) + i \frac{\sin(\pi z \Omega)}{\Omega} M \right] \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Since

$$M \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} s \\ 1/\xi \end{pmatrix},$$

the beam amplitudes are

$$u_0(z) = \cos(\pi z \Omega) + i \frac{s}{\Omega} \sin(\pi z \Omega),$$

$$u_{\mathbf{g}}(z) = i \frac{1/\xi}{\Omega} \sin(\pi z \Omega).$$

### 3.5 Dynamical intensity

The diffracted intensity is

$$I_{\text{dyn}}(z, s) \propto |u_{\mathbf{g}}(z)|^2,$$

so

$$I_{\text{dyn}}(z, s) \propto \frac{1/\xi^2}{\Omega^2} \sin^2(\pi z \Omega).$$

Using

$$\Omega^2 = s^2 + \frac{1}{\xi^2},$$

and then setting  $z = t$ , we obtain

$$I_{\text{dyn}}(t, s) \propto \frac{1/\xi^2}{s^2 + 1/\xi^2} \sin^2 \left( \pi t \sqrt{s^2 + \frac{1}{\xi^2}} \right).$$

Multiplying numerator and denominator by  $\xi^2$  gives

$$I_{\text{dyn}}(t, s) \propto \frac{\sin^2 \left[ \left( \frac{\pi t}{\xi} \right) \sqrt{1 + (\xi s)^2} \right]}{1 + (\xi s)^2}.$$

### 3.6 Equivalent sinc form of the dynamical expression

Let

$$w = \xi s.$$

Then

$$I_{\text{dyn}}(w) \propto \frac{\sin^2 \left[ \left( \frac{\pi t}{\xi} \right) \sqrt{1 + w^2} \right]}{1 + w^2}.$$

Using (1) with

$$x = \frac{t}{\xi} \sqrt{1 + w^2},$$

one finds

$$\sin^2 \left[ \left( \frac{\pi t}{\xi} \right) \sqrt{1 + w^2} \right] = \pi^2 \left( \frac{t}{\xi} \right)^2 (1 + w^2) \text{sinc}^2 \left( \frac{t}{\xi} \sqrt{1 + w^2} \right).$$

Therefore,

$$I_{\text{dyn}}(w) \propto \pi^2 \left( \frac{t}{\xi} \right)^2 \text{sinc}^2 \left( \frac{t}{\xi} \sqrt{1 + w^2} \right).$$

### 3.7 Comparison dynamical versus kinematical expressions

With  $w = \xi s$ , the two rocking-curve expressions are

$$I_{\text{kin}}(w) \propto \pi^2 \left(\frac{t}{\xi}\right)^2 \text{sinc}^2\left(\frac{t}{\xi}w\right),$$

and

$$I_{\text{dyn}}(w) \propto \pi^2 \left(\frac{t}{\xi}\right)^2 \text{sinc}^2\left(\frac{t}{\xi}\sqrt{1+w^2}\right).$$

Thus the kinematical result depends linearly on  $w$  in the sinc argument, whereas the simplified two-beam dynamical result replaces  $w$  by  $\sqrt{1+w^2}$ , reflecting the coupled-wave nature of the propagation problem. These expressions are not identical in general. In the weak-coupling regime  $t/\xi \ll 1$ , at fixed excitation error  $s$ , the dynamical expression reduces to the kinematical line shape up to the same normalization convention. The condition of fixed  $s$  matters because fixing  $w = \xi s$  while varying  $\xi$  corresponds to a different physical detuning regime.

## 4 A simple nontrivial absorption model

A simple way to obtain a nontrivial absorptive modification of the dynamical rocking curve is to allow the two-beam coupling itself to become complex. To avoid repeating the previous matrix-exponential derivation, we reuse the non-absorbing solution and apply the substitution

$$\frac{1}{\xi} \longrightarrow \gamma, \quad \gamma = \frac{1}{\xi} + i\frac{1}{\xi'},$$

where  $\xi'$  is an absorptive length scale.

With the same entrance condition  $u_0(0) = 1$ ,  $u_g(0) = 0$ , define

$$\Omega_{\text{abs}} = \sqrt{s^2 + \gamma^2}.$$

Then, by direct analogy with the non-absorbing two-beam result,

$$u_0(z) = \cos(\pi z \Omega_{\text{abs}}) + i \frac{s}{\Omega_{\text{abs}}} \sin(\pi z \Omega_{\text{abs}}),$$

$$u_g(z) = i \frac{\gamma}{\Omega_{\text{abs}}} \sin(\pi z \Omega_{\text{abs}}).$$

Hence

$$I_{\text{dyn,abs}}(z, s) \propto \left| \frac{\gamma}{\Omega_{\text{abs}}} \sin(\pi z \Omega_{\text{abs}}) \right|^2,$$

and at  $z = t$ ,

$$I_{\text{dyn,abs}}(t, s) \propto \left| \frac{\gamma}{\sqrt{s^2 + \gamma^2}} \sin\left(\pi t \sqrt{s^2 + \gamma^2}\right) \right|^2.$$

Using (1),

$$I_{\text{dyn,abs}}(t, s) \propto \pi^2 t^2 |\gamma|^2 \left| \text{sinc}\left(t \sqrt{s^2 + \gamma^2}\right) \right|^2.$$

Because  $\gamma$  is complex,  $\Omega_{\text{abs}}$  is generally complex as well, so absorption changes the rocking-curve shape itself (not only its overall scale).

## 4.1 The kinematical absorbing case

In the kinematical case, a simple phenomenological way to include absorption is through a damped thickness integral, for example

$$u_{\mathbf{g}}^{(\text{kin,abs})}(t, s) \propto \frac{1}{\xi} \int_0^t e^{2\pi i s z} e^{-\mu(t-z)} dz.$$

Evaluating the integral gives

$$u_{\mathbf{g}}^{(\text{kin,abs})}(t, s) \propto \frac{1}{\xi} \frac{e^{2\pi i s t} - e^{-\mu t}}{2\pi i s + \mu},$$

and hence

$$I_{\text{kin,abs}}(t, s) \propto \frac{\pi^2}{\xi^2} \left| \frac{e^{2\pi i s t} - e^{-\mu t}}{2\pi i s + \mu} \right|^2.$$

Thus, in this next-simplest absorption treatment, the kinematical and dynamical cases are modified in fundamentally different ways:

- in the kinematical case, absorption enters through the thickness integral,
- in the dynamical case, absorption enters through a complex coupled-wave propagation constant.

This is why the absorptive modification is no longer just a common overall prefactor.

## 4.2 Final forms

The absorbing simplified two-beam dynamical rocking curve is

$$I_{\text{dyn,abs}}(t, s) \propto \pi^2 t^2 |\gamma|^2 \left| \text{sinc}\left(t\sqrt{s^2 + \gamma^2}\right) \right|^2, \quad \gamma = \frac{1}{\xi} + i\frac{1}{\xi'}.$$

This reduces to the non-absorbing dynamical result when

$$\frac{1}{\xi'} \rightarrow 0,$$

since then  $\gamma \rightarrow 1/\xi$ .

The absorbing kinematical expression was

$$I_{\text{kin,abs}}(t, s) \propto \frac{\pi^2}{\xi^2} \left| \frac{e^{2\pi i s t} - e^{-\mu t}}{2\pi i s + \mu} \right|^2.$$

In the limit  $\mu \rightarrow 0$ ,

$$I_{\text{kin,abs}}(t, s) \rightarrow \frac{\pi^2}{\xi^2} \left| \frac{e^{2\pi i s t} - 1}{2\pi i s} \right|^2.$$

To recover the sinc form, rewrite the numerator as

$$e^{2\pi i s t} - 1 = e^{\pi i s t} (e^{\pi i s t} - e^{-\pi i s t}) = 2i e^{\pi i s t} \sin(\pi s t).$$

Hence

$$\frac{e^{2\pi i s t} - 1}{2\pi i s} = e^{\pi i s t} \frac{\sin(\pi s t)}{\pi s} = t e^{\pi i s t} \frac{\sin(\pi s t)}{\pi s t} = t e^{\pi i s t} \text{sinc}(st).$$

Therefore

$$\left| \frac{e^{2\pi i s t} - 1}{2\pi i s} \right|^2 = t^2 \text{sinc}^2(st),$$

since  $|e^{\pi i s t}|^2 = 1$ .

Thus

$$I_{\text{kin,abs}}(t, s) \rightarrow \pi^2 \left( \frac{t}{\xi} \right)^2 \text{sinc}^2(st),$$

which is exactly the non-absorbing kinematical result.



## 5 Phenomenological identification of the absorption scales

The absorption parameters introduced in the kinematical and dynamical treatments are in general not identical at the level of the underlying derivation. In the kinematical model, absorption was represented by the damping coefficient  $\mu$ , while in the dynamical model it was represented by the absorptive inverse length  $1/\xi'$  entering the complex coupling constant.

For comparison purposes, however, it is convenient to relate the two through a phenomenological proportionality

$$\frac{1}{\xi'} = C\mu,$$

where  $C$  is a dimensionless model-dependent scale factor. It should not be interpreted as a universal constant that enforces one model to be always more or less absorption-sensitive than the other across all  $t, \xi, s, \mu$ ; that comparison depends on parameter regime.

With this identification, both absorption parameters vanish simultaneously when

$$\mu \rightarrow 0,$$

so that both the kinematical and dynamical absorbing expressions reduce to their corresponding non-absorbing forms.

### 5.1 Final forms with a common absorption parameter

In the absorbing dynamical model, the complex coupling constant was defined as

$$\gamma = \frac{1}{\xi} + i\frac{1}{\xi'}.$$

Using

$$\frac{1}{\xi'} = C\mu,$$

this becomes

$$\gamma = \frac{1}{\xi} + iC\mu.$$

The absorbing kinematical rocking curve is

$$I_{\text{kin,abs}}(t, s) \propto \frac{\pi^2}{\xi^2} \left| \frac{e^{2\pi i s t} - e^{-\mu t}}{2\pi i s + \mu} \right|^2,$$

and the absorbing dynamical rocking curve becomes

$$I_{\text{dyn,abs}}(t, s) \propto \pi^2 t^2 \left| \frac{1}{\xi} + iC\mu \right|^2 \left| \text{sinc} \left( t \sqrt{s^2 + \left( \frac{1}{\xi} + iC\mu \right)^2} \right) \right|^2.$$

Both expressions reduce to their corresponding non-absorbing forms in the limit

$$\mu \rightarrow 0.$$